Uses of Life Table

- Actuaries compute rate of premiums for persons of different age groups using Life Table.
- It helps to assess the accuracy of census figures, death and birth registrations.
- It helps to evaluate the impact of family planning on population growth.
- It enables to assess the increase in the life span due to new scientific inventions, sophisticated medical treatments and improved living conditions.
- Estimates of migration can be made from Life Table.

Construction of Life Table

Construction of Life Table begins with a ***cohort*** population. The following are the standard set of components of a Life Table:

(i) Age (x)
(ii) Survivorship function
(iii) Number of deaths in the age interval (x, x+1)
(iv) Probability for a person surviving at the age x to die before x+1 years
(v) Probability for a person aged x years to survive upto x+1 years
(vi) Number of persons lived in aggregate in the age interval (x,x+1)
(vii) Number of years lived by the ***cohort*** at and above the age x years
(viii) Expectation of life.

These components are described below with their respective notations and formula required to compute each of them.

(i) x : Age, in years

(ii) $l(x)$: Number of survivors at the exact age of x years.

For instance, $l(25)$ denotes the number of persons who survive to the moment of attaining age 25 years. Hence, $l(x)$ is a decreasing function of x.

$l(0)$ is called **radix** of the Life Table or ***cohort***, which is conventionally taken as 1,00,000.

(iii) $d(x)$: Number of persons among $l(x)$ persons who die before reaching the age x years.

i.e., $d(x) = l(x) - l(x+1)$

(iv) $q(x)$: Probability for a person surviving at the age x will die before x+1 years.

i.e., $q(x) = \frac{d(x)}{l(x)}$

It is the proportion of persons dying between the ages of x and x +1 to the number of persons surviving at the age of x years, *i.e.*, at the beginning of the corresponding age interval.

(v) $p(x)$: Probability for a person aged x years to survive up to x+1 years

i.e., $p(x) = 1 - q(x)$, or equivalently, $p(x) = \frac{l(x+1)}{l(x)}$

It is the proportion of persons surviving up to (x + 1) years to the number of persons at the age of x years.

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